AI BASED ADAPTIVE VR MEDITATION SYSTEM

Project ID: R26-DS-006

Project Proposal Report

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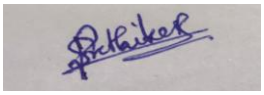
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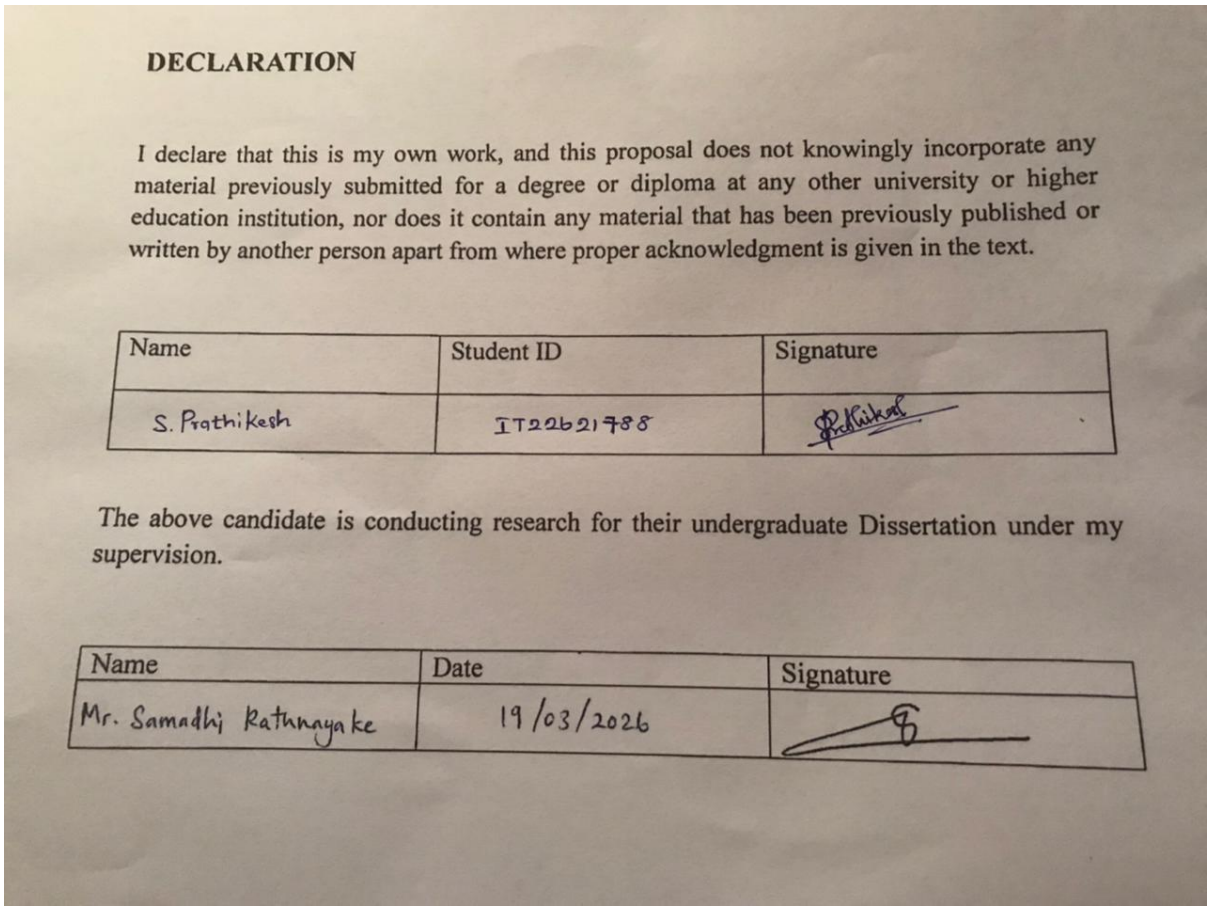
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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Date	Signature
S. Prathikesh	2026.03.19	



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ABSTRACT

This research proposes the development of an AI-based stress-aware Virtual Reality (VR) meditation system to improve stress assessment, intervention monitoring, and adaptive wellness support through a multimodal intelligent framework. Stress has become a major concern in modern life, creating a strong need for accessible, personalized, and technology-driven stress-management solutions. Although existing VR-based relaxation systems provide immersive therapeutic experiences, many rely primarily on physiological sensing and offer limited ability to verify whether a meditation session has effectively reduced the user's stress. This reveals a research gap in developing an integrated framework that combines physiological stress inference, voice-based stress assessment, and session-level evaluation within a unified VR intervention environment.

To address this gap, the proposed system integrates real-time physiological signal acquisition, HRV-based baseline stress estimation, multilingual voice-based stress assessment, and anomaly-based session validation. Physiological signals are used to estimate baseline stress, while voice recordings collected before and after the meditation session are analyzed to identify changes in acoustic stress patterns. The current implementation targets English, with Sinhala under active development and Tamil planned as a future extension. A fine-tuned XLSR-53 model is used for acoustic stress scoring, and baseline voice-derived stress is compared with HRV-based stress outputs through a cross-modal validation mechanism. In addition, an autoencoder-based anomaly detection module is incorporated to identify unusual or ineffective meditation sessions.

The expected outcome is a reliable and adaptive stress-aware VR system that enhances stress evaluation, improves session monitoring, and supports personalized digital wellness interventions. Socially, the system can contribute to accessible mental wellness support, while commercially it has potential applications in digital therapeutics, wellness platforms, and intelligent healthcare solutions.

Keywords - VR meditation, HRV-based stress estimation, multilingual voice-based stress assessment, cross-modal validation, anomaly detection.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to [Mr. Samadhi Rathnayake] for the invaluable guidance, constructive feedback, and continuous support provided throughout the development of this research proposal. Their expertise and encouragement were instrumental in shaping the direction and quality of this work.

I also extend my appreciation to the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary resources, academic environment, and opportunities that supported this research.

Finally, I am deeply grateful to my family and friends for their unwavering encouragement, patience, and motivation throughout this journey.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
VR	Virtual Reality
XLSR	Cross-Lingual Speech Representations
WAV	Waveform Audio File Format
RMSSD	Root Mean Square of Successive Differences
ML	Machine Learning
SDNN	Standard Deviation of NN Intervals

UI	User Interface
HRV	Heart Rate Variability
API	Application Programming Interface
DRS	Design Science Research

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Appendix A: Stakeholder Requirements Survey Instrument

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1. INTRODUCTION

1.1 Background and Context

1.1.1 The Real-World Problem

Stress has become one of the most pervasive challenges affecting mental health and quality of life in modern society. Across academic, professional, and personal environments, unmanaged stress contributes to reduced cognitive performance, emotional instability, and long-term physical health consequences. The World Health Organization has identified stress-related mental health conditions as a leading contributor to disability and reduced quality of life globally [11]. In Sri Lanka, this concern is particularly evident among university students and working populations, where psychological distress and mental health difficulties have been documented at notable levels [12]. Despite growing awareness, access to professional mental health support and stress-relief interventions remains limited for many individuals due to cost, availability, and social stigma.

1.1.2 Who Is Affected and Why It Matters

The impact of unmanaged stress extends across all segments of society. University students face academic pressure, financial uncertainty, and social challenges that collectively contribute to elevated stress levels. Working professionals experience workplace demands, deadlines, and performance expectations that can lead to chronic stress and burnout. In both cases, the inability to access timely and effective stress-relief support can result in long-term mental and physical health deterioration.

Traditional stress-relief approaches such as in-person meditation centers, therapy sessions, and wellness programs, while effective, are not always accessible. Geographic limitations, financial constraints, and time pressures prevent many individuals from regularly attending such services. This creates a clear need for an intelligent, accessible, and cost-effective alternative that can be used at home without specialized equipment or professional supervision.

1.1.3 Statistical Evidence

Anxiety disorders are the most common mental illness worldwide, affecting approximately 301 million people globally, with stress-related conditions representing a significant and growing public health concern that affects hundreds of millions of people across all age groups and demographics [11]. The COVID-19 pandemic alone triggered a 25% increase in the global prevalence of anxiety and depression, underscoring the urgent and growing demand for accessible mental health interventions [14]. Depression and anxiety are estimated to cost the global economy approximately one trillion US dollars each year in lost productivity, highlighting the significant societal and economic burden of unmanaged stress [11]. In Sri Lanka, stress-related mental health challenges are particularly prevalent among university students, where psychological distress has been documented at notable levels, pointing to the need for accessible and low-cost stress-relief solutions [12].

With regard to technology-based interventions, VR-based mindfulness and meditation systems have demonstrated measurable physiological and psychological benefits in controlled studies. A randomized controlled study involving 87 participants found that a VR-supported Heart Rate Variability biofeedback intervention significantly improved stress indicators and stress-related symptoms, achieving good to very good user experience outcomes [13]. A further study confirmed that VR-based setups significantly reduced perceived stress and increased Heart Rate Variability, with head-mounted display delivery yielding greater improvements in cardiac coherence than conventional

screen-based approaches [1]. Despite these promising outcomes, existing systems continue to rely on questionnaire-based self-report or laboratory-grade physiological equipment to measure stress reduction, leaving a clear gap for automated, voice-based session evaluation within consumer-grade VR wellness systems.

1.1.4 Analysis of Existing Solutions and Their Limitations

Current approaches to stress management and digital wellness can be grouped into three main categories, each with notable limitations when considered in the context of intelligent, session-level stress evaluation.

The first category comprises in-person meditation and therapy services. While these approaches are clinically effective, they require physical attendance, professional scheduling, and consistent financial investment. For many individuals — particularly university students and working professionals in Sri Lanka — these barriers make regular participation impractical, highlighting the need for a more accessible alternative.

The second category includes wearable-based physiological monitoring solutions such as consumer heart rate trackers and HRV-enabled smartwatches. Although these devices are useful for capturing autonomic stress responses in real time, they rely exclusively on body-level signals and do not incorporate behavioral indicators such as speech. As a result, they are unable to assess cognitive or emotional stress that may not manifest strongly in physiological readings alone, and they provide no mechanism for evaluating whether a meditation intervention has produced a measurable change in the user's stress state [2], [7].

The third category covers commercially available VR meditation platforms, most notably TRIPP, which has delivered over 14 million sessions and has been incorporated into clinical research studies [1]. Despite its commercial success, TRIPP evaluates intervention effectiveness through self-report questionnaires or laboratory-grade equipment rather than automated signal-based analysis. All reviewed studies using such platforms were conducted under controlled experimental conditions, meaning no existing VR system currently accounts for the real-world variability users face at home, nor does any system integrate voice-based stress assessment, cross-modal validation, or automated session outcome evaluation into a unified consumer-grade framework.

1.1.5 Proposed Solution Overview

This research forms part of a broader AI-based stress-aware Virtual Reality meditation system, contributing specifically to the voice-based stress assessment and session validation component. The proposed module addresses the limitations identified in existing solutions by introducing a non-invasive, speech-based approach to measuring and validating stress relief outcomes at the session level.

The module operates in four stages. In the first stage, the surrounding acoustic environment is briefly assessed through the desktop microphone before the session begins, and the user is notified if ambient noise levels are unsuitable for effective VR meditation. In the second stage, a short voice recording is collected from the user before and after the VR meditation session using a standard desktop

microphone. These recordings are processed through a fine-tuned XLSR-53 cross-lingual speech foundation model to generate acoustic stress scores that reflect the user's stress state based on vocal characteristics such as pitch variation, speech rate, and vocal stability. The current implementation supports English and Sinhala, with Tamil considered as a future extension [5], [8], [9].

In the third stage, the voice-derived stress estimates are compared against HRV-based physiological stress outputs produced by the stress inference module through cross-modal validation, enabling the system to detect cases where physiological and cognitive stress indicators diverge [2], [7]. In the fourth stage, an Autoencoder-based anomaly detection mechanism analyzes session-level stress patterns to identify unusual or ineffective meditation sessions that deviate from expected stress reduction outcomes [6].

By combining these four functions into a unified workflow, the proposed module strengthens session-level stress evaluation and supports the development of a more reliable, adaptive, and intelligent stress management system.

1.1.6 Relevant SDG Alignment

SDG	Goal	Alignment Rationale
SDG 3	Good Health and Well-being	Directly supports mental health and stress reduction by providing an accessible, AI-powered VR meditation system for individuals who cannot access traditional wellness services, contributing to improved psychological well-being across the Sri Lankan population.
SDG 10	Reduced Inequalities	Supports multilingual stress assessment in English and Sinhala, making the system accessible to a broader and more diverse Sri Lankan population regardless of language background, with Tamil considered as a future extension.
SDG 9	Industry, Innovation and Infrastructure	Introduces an innovative AI-driven framework combining cross-lingual speech foundation models, cross-modal stress validation, and Autoencoder-based anomaly detection within a consumer-grade VR wellness system — advancing the frontier of intelligent digital health infrastructure.
SDG 4	Quality Education	Directly addresses stress among university students — a key target population — by providing an accessible, intelligent, and self-managed stress-relief tool that supports academic performance and student well-being.

1.1.7 Research Cluster Alignment

This research fits within the **Intelligent Systems and Data Science** research cluster, particularly in the areas of applied machine learning, speech-based affective computing, multimodal signal fusion, and AI-driven digital health applications. The use of the XLSR-53 cross-lingual speech foundation model for acoustic stress assessment and the Autoencoder-based anomaly detection mechanism for session evaluation both represent direct contributions to this cluster through novel applications of deep learning and unsupervised anomaly detection in a therapeutic context.

The research also connects to the Human-Computer Interaction **and** Digital Wellbeing cluster through its focus on Virtual Reality-based stress relief interventions, user-centered system design, and the development of non-invasive, consumer-grade tools for mental health support. By integrating voice-based behavioral signals with physiological biofeedback within a VR environment, the proposed module advances the intersection of affective computing, digital wellness, and adaptive human-computer interaction.

1.2 Literature Review

1.2.1 Voice-Based Stress Detection and Acoustic Stress Analysis

Speech has increasingly been recognized as a meaningful and non-invasive indicator of psychological stress, with acoustic and prosodic features of voice shown to respond specifically to physiological stress activation. Research by Kappen et al. [2] examined how acoustic features including fundamental frequency, jitter, and speaking rate reflect physiological stress responses across multiple psychosocial stressor paradigms, finding that these vocal characteristics respond specifically to stress-inducing conditions rather than general negative affect. This distinction is important as it positions voice as an objective physiological signal rather than a purely subjective behavioral indicator. A systematic review examining 38 independent studies confirmed that prosodic speech features demonstrated the most consistent association with stress and cognitive load, although reported findings varied depending on experimental context and recording environment [7]. Together, these findings establish a strong foundation for using voice as a primary input modality for stress assessment within the proposed system.

Beyond single-point stress classification, recent work has explored the potential of acoustic features to track psychological change over time. A systematic review of voice-based applications in mental health confirmed that speech-derived acoustic features can reflect changes in psychological state that are not always detectable through conventional self-report instruments alone [10]. This is particularly relevant to the proposed module, where the goal extends beyond detecting initial stress to evaluating whether the VR meditation session produced a measurable vocal stress reduction across the intervention.

1.2.2 Multimodal Stress Recognition and Cross-Modal Validation

A growing body of research has demonstrated that combining multiple signal sources produces more reliable stress recognition outcomes than relying on any single modality alone. The MuSE dataset [3] and the WESAD dataset [4] represent two of the most widely used multimodal resources in stress and affect detection research, collectively demonstrating that physiological and behavioral signals capture distinct and complementary dimensions of stress. WESAD includes wrist and chest-worn physiological sensors alongside self-report labels, providing a benchmark for evaluating the complementary value of different signal types [4].

With specific regard to combining voice and physiological signals, a systematic review and meta-analysis focusing on vocal fundamental frequency as a stress biomarker found that voice-only systems face significant challenges in differentiating stress from other emotionally aroused states, and that incorporating complementary physiological measures such as HRV is likely necessary for reliable

discrimination [10]. This finding directly supports the cross-modal validation approach proposed in this research, where voice-derived acoustic stress scores are compared against HRV-based physiological stress estimates to assess agreement and detect cases where cognitive and physiological stress indicators diverge. A randomized controlled study further confirmed that VR-supported HRV biofeedback interventions can produce significant improvements in stress-related physiological indicators, establishing the validity of HRV as a reference signal for cross-modal comparison [13].

1.2.3 Cross-Lingual Speech Models, Low-Resource Languages, and Anomaly Detection

The selection of the XLSR-53 architecture for voice-based stress assessment is motivated by the practical requirement to support Sinhala, a low-resource language with limited labelled emotional speech data. XLSR was developed as a self-supervised cross-lingual speech model pretrained across 53 languages simultaneously, demonstrating strong transferability to downstream speech classification tasks even in settings where labelled data is scarce [5]. This cross-lingual capability makes XLSR-53 particularly appropriate for the current implementation, where sufficient labelled Sinhala stress speech data is not readily available. The feasibility of this approach is further supported by a growing body of Sinhala speech processing research, including the development of a dedicated Sinhala emotion dataset [8] and the adaptation of cross-lingual speech emotion recognition models for the Sinhala language [9], collectively confirming that multilingual speech-based affective computing in this linguistic context is both practical and achievable.

With regard to session-level outcome evaluation, unsupervised anomaly detection offers a principled and practical approach to identifying ineffective or unusual meditation sessions without requiring manually labelled examples of failure. Recent research demonstrated that Autoencoder architectures optimized through latent space manipulation can reliably identify deviations from expected patterns by comparing the reconstruction accuracy of new inputs against the normal distribution the model was trained on [6]. Applied to the proposed system, this approach enables an Autoencoder trained on typical stress reduction patterns to automatically flag sessions where the expected vocal stress change does not occur — supporting adaptive system response without the need for labelled anomaly data.

1.3 Research Gap

1.3.1 What Is Missing from Existing Research

A close review of the literature reveals four connected gaps that collectively motivate the proposed research. First, existing VR meditation platforms including commercially deployed systems such as TRIPP evaluate intervention effectiveness through self-report questionnaires or laboratory-grade equipment rather than through automated, continuous signal-based analysis [1]. No existing VR meditation system incorporates a real-time, non-invasive voice-based mechanism for measuring whether a session has reduced the user's stress level. Second, while voice-based stress detection and physiological stress monitoring have each been studied independently, no published system has combined voice-derived acoustic stress scores with HRV-based physiological stress estimates for cross-modal validation within a VR meditation context [2], [7], [10]. Third, existing voice stress detection systems are developed and evaluated exclusively on English speech datasets, with no published work applying cross-lingual speech foundation models to multilingual stress assessment in a VR therapeutic setting that includes Sinhala or other South Asian languages [5], [8], [9]. Fourth, session-level anomaly detection — the automated identification of unusual or ineffective meditation

outcomes — has not been addressed in any existing VR wellness system, leaving no mechanism to detect when an intervention fails to produce the expected stress reduction pattern [6].

1.3.2 Novelty, Creativity, and Evidence-Based Justification

At the signal level, this research introduces voice as an independent and complementary stress assessment modality within a VR meditation system. Unlike physiological signals such as HRV, which primarily reflect autonomic nervous system responses, voice captures cognitive and emotional dimensions of stress that may persist even after the body has physically relaxed [2], [10]. This distinction is practically significant — a user may show normalized HRV readings while still experiencing cognitive stress that manifests through vocal tension, elevated pitch, and disrupted speech rhythm. No existing VR meditation system exploits this distinction, meaning that current systems have no mechanism to detect cases where the body relaxes but the mind remains stressed. The proposed module directly addresses this gap by capturing both physiological and acoustic stress dimensions and comparing them explicitly.

At the method level, the proposed approach introduces three technically distinct contributions. First, the application of a fine-tuned XLSR-53 cross-lingual speech foundation model for acoustic stress scoring represents a significant methodological advancement over existing voice stress systems, which have relied on basic acoustic features such as MFCCs and traditional classifiers rather than deep self-supervised speech representations [5], [7]. Second, the cross-modal validation framework — comparing voice-derived stress scores against HRV-based physiological estimates — provides a principled mechanism for detecting cognitive-physiological stress mismatches that no existing single-modality system can identify [2], [10]. Third, the application of an Autoencoder-based anomaly detection mechanism to session-level stress reduction patterns enables the system to automatically identify ineffective meditation sessions without requiring manually labelled failure examples — a capability that does not exist in any published VR wellness system [6].

At the system level, the proposed module is the first to integrate voice-based acoustic stress assessment, cross-modal physiological validation, multilingual support for English and Sinhala, and Autoencoder-based session anomaly detection within a single consumer-grade VR meditation framework. All reviewed VR meditation studies were conducted in controlled laboratory environments — none have deployed voice-based session evaluation in a real-world home setting where ambient noise, language diversity, and session variability are practical concerns [1].

The feasibility of this work is supported in three ways. First, fine-tuned cross-lingual speech models based on the Wav2Vec2 architecture have demonstrated consistent speech emotion recognition accuracy above 75% across multiple languages in published benchmarks, establishing a realistic and achievable performance baseline for the proposed implementation [5], [9]. Second, multimodal stress research using established datasets including WESAD [4] and MuSE [3] has confirmed that combining voice and physiological signals produces more reliable and comprehensive stress assessment than either modality alone, directly validating the cross-modal approach adopted here. Third, since no existing VR meditation system incorporates voice-based stress assessment, cross-modal validation, or session-level anomaly detection, the research gap is clearly defined, practically motivated, and supported by a consistent body of literature — confirming that the opportunity for meaningful and original contribution is significant.

Research Gap	Existing Limitation	This Research's Contribution
Voice-Based Session Evaluation	No existing VR meditation system measures stress from voice before or after the session — effectiveness is assessed only through questionnaires or lab-grade equipment	Pre-session and post-session acoustic stress scoring using a fine-tuned XLSR-53 cross-lingual speech foundation model
Cross-Modal Stress Validation	Voice-based and physiological stress signals have only been studied in isolation — no system compares them within a VR meditation context	Cross-modal validation framework comparing voice-derived acoustic stress scores against HRV-based physiological stress estimates to detect cognitive-physiological mismatches
Multilingual Support	All existing voice stress detection systems are developed and evaluated on English speech only — no system supports Sinhala or other South Asian languages	English and Sinhala language support through XLSR-53 cross-lingual pretraining with Tamil considered as a future extension
Session Anomaly Detection	No VR wellness system includes an automated mechanism to detect ineffective or unusual meditation sessions	Autoencoder-based anomaly detection trained on normal stress reduction patterns to automatically flag sessions that deviate from expected outcomes

2. OBJECTIVES

2.1 Main Objective

To design, develop, and evaluate a multilingual voice-based stress assessment and cross-modal session validation module for an AI-based stress-aware Virtual Reality meditation system. The module will collect pre-session and post-session voice recordings in English and Sinhala, apply a fine-tuned XLSR-53 cross-lingual speech foundation model to generate acoustic stress scores, validate voice-derived stress estimates against HRV-based physiological outputs through cross-modal comparison, and apply an Autoencoder-based anomaly detection mechanism to identify unusual or ineffective meditation sessions. The target is to achieve a minimum acoustic stress classification accuracy of 75% across English and Sinhala speakers on the evaluation dataset.

2.2 Specific Objectives

1. To develop a pre-session ambient environment assessment mechanism that analyzes surrounding acoustic noise levels through a standard desktop microphone and determines whether the user's environment is suitable for effective VR meditation, providing an appropriate readiness notification before the session begins.
2. To collect, organize, and preprocess pre-session and post-session voice recordings in English and Sinhala, applying audio preprocessing steps including resampling to 16kHz, silence

trimming, amplitude normalization, and waveform formatting to prepare high-quality input data for acoustic stress analysis and model training.

3. To fine-tune the XLSR-53 cross-lingual speech foundation model on a custom stress-labelled dataset combining established English speech emotion datasets including RAVDESS and CREMA-D with locally collected Sinhala voice recordings, achieving a minimum stress classification accuracy of 75% across both supported languages on the held-out evaluation set.
4. To develop a pre-session and post-session stress comparison mechanism that analyzes voice-derived acoustic stress scores before and after the VR meditation session to identify stress-related vocal changes and determine whether a meaningful stress reduction occurred across the intervention.
5. To implement a cross-modal validation framework that compares voice-derived acoustic stress estimates against HRV-based physiological stress outputs produced by the stress inference module, assessing agreement between the two modalities and detecting cases where cognitive and physiological stress indicators diverge.
6. To design and train an Autoencoder-based anomaly detection model from scratch using session-level acoustic stress features, enabling the system to automatically identify unusual or ineffective meditation sessions by measuring deviation from learned normal stress reduction patterns.
7. To design the overall module in a scalable and modular manner so that, beyond the current English and Sinhala implementation, the framework can be extended to support Tamil and additional languages in future work without requiring fundamental architectural changes.

3. METHODOLOGY

3.1 Research Approach

This research follows the Design Science Research (DSR) methodology, which is centered on the design, development, and rigorous evaluation of innovative technical artifacts to address clearly identified real-world problems. DSR was selected because it provides a structured and reproducible framework for building and assessing AI-driven systems in applied domains, making it particularly well suited for the development of a voice-based stress assessment module within a VR meditation context. The research will proceed through four DSR phases: Requirements Analysis and Module Design, System Implementation and Model Development, Empirical Evaluation and Validation, and Reflection and Future Extension Planning.

3.2 Technical Methods and Algorithms

3.2.1 Layer 1 — Ambient Environment Assessment

Prior to each VR meditation session, the surrounding acoustic environment is analyzed using the desktop microphone to determine whether ambient noise levels are suitable for effective stress relief. A five-second audio sample is recorded and processed to extract the root mean square energy level and zero crossing rate, which together serve as indicators of background noise intensity. If the computed noise level exceeds a predefined threshold calibrated through preliminary experiments, the system displays a readiness notification advising the user to move to a quieter environment before proceeding.

This step does not involve a machine learning model — it uses signal-level acoustic analysis to produce a binary readiness status that is logged alongside the session record.

3.2.2 Layer 2 — Voice-Based Acoustic Stress Assessment Using XLSR-53

The core stress assessment pipeline is built upon the XLSR-53 cross-lingual speech foundation model, which was pretrained in a self-supervised manner across 53 languages using raw audio waveforms. This pretraining approach enables the model to learn deep acoustic representations that capture language-independent phonetic, prosodic, and paralinguistic patterns — including stress-related vocal characteristics such as pitch variation, speech rate, jitter, and vocal energy — without requiring manually labelled data during pretraining.

For the purposes of this research, XLSR-53 is fine-tuned for binary stress classification using a custom dataset combining established English speech emotion corpora including RAVDESS and CREMA-D with locally collected Sinhala voice recordings. The fine-tuning process involves appending a classification head consisting of a mean pooling layer followed by two fully connected layers and a softmax output over two classes — stressed and not stressed — to the XLSR-53 encoder. During fine-tuning, the classification head is trained while the encoder layers are selectively unfrozen based on dataset size and validation performance, preserving the general cross-lingual representations learned during pretraining while adapting the model to the stress classification task.

The input to the model is a raw audio waveform resampled to 16kHz, preprocessed through silence trimming, amplitude normalization, and fixed-length padding or truncation to a 30-second window. The model outputs a stress probability score between 0 and 1, which is scaled to a 0 to 10 acoustic stress score for interpretability. This scoring process is applied independently to the pre-session and post-session voice recordings, producing two comparable stress scores per session.

3.2.3 Layer 3 — Pre-Session and Post-Session Stress Change Analysis

Following the generation of acoustic stress scores for both the pre-session and post-session recordings, a stress change analysis is performed to determine whether the VR meditation session produced a measurable vocal stress reduction. The absolute and relative differences between the two scores are computed, and a reduction threshold determined through preliminary validation experiments is applied to classify the session outcome as stress-reduced, unchanged, or stress-increased. This analysis produces a stress change indicator that is stored in the session log and made available for cross-modal validation and anomaly detection.

3.2.4 Layer 4 — Cross-Modal Validation Framework

The cross-modal validation framework compares the voice-derived acoustic stress score against the HRV-based physiological stress estimate produced by the stress inference module of the broader system. For each session, both the pre-session and post-session scores from each modality are compared using an absolute difference threshold to determine whether the two signals agree or divergence. Agreement indicates that both physiological and acoustic indicators consistently reflect the same stress state, strengthening confidence in the session of outcome assessment. Divergence — where the physiological signal suggests relaxation, but the acoustic signal remains elevated — is flagged as a cognitive-physiological mismatch, indicating that the user's body may have relaxed while

cognitive or emotional stress persisted. The agreement status, divergence flag, and modality-specific scores are all recorded in the session of output for system-level interpretation.

3.2.5 Layer 5 — Autoencoder-Based Session Anomaly Detection

Session anomaly detection is implemented using an Autoencoder architecture designed and trained entirely from scratch. The Autoencoder is trained exclusively on session records from effective meditation sessions — defined as sessions where both the acoustic and physiological stress scores show a meaningful reduction — enabling it to learn a compact latent representation of normal stress relief patterns. During inference, each new session record is passed through the trained Autoencoder, and the reconstruction error — measured as mean squared error between the input and the reconstructed output — is computed. Sessions with reconstruction error above a threshold determined during validation are flagged as anomalous, indicating that the session did not follow the expected stress reduction pattern.

The input feature vector for the Autoencoder includes the pre-session acoustic stress score, post-session acoustic stress score, acoustic stress change value, pre-session physiological stress score, post-session physiological stress score, physiological stress change value, and cross-modal agreement status. The architecture consists of an encoder with two fully connected layers reducing the input to a compact bottleneck representation, followed by a symmetric decoder that reconstructs the original feature vector. The model is trained using mean squared error loss and the Adam optimizer, running until convergence with early stopping based on validation reconstruction error.

3.3 Tools and Platforms with Justification

Tool / Platform	Role	Justification
Python 3.11	Primary development language	Dominant language for machine learning and speech processing research with extensive library support
XLSR-53 via Hugging Face Transformers	Cross-lingual speech foundation model	foundation modelSelf-supervised pretraining across 53 languages enables strong transfer to low-resource Sinhala stress classification
PyTorch	Model development and fine-tuning framework	Industry-standard deep learning framework with flexible support for custom model architectures and fine-tuning pipelines
Librosa	Audio feature extraction and preprocessing	Well-established library for audio signal processing including resampling, silence trimming, and spectral feature extraction
Torchaudio	Audio loading and transformation	PyTorch-native audio processing library providing efficient waveform loading and augmentation
SoundDevice / PyAudio	Real-time microphone recording	Lightweight libraries for capturing desktop microphone input for pre-session and post-session voice recording
OpenAI Whisper	Speech transcription for preference extraction	Multilingual automatic speech recognition model supporting Sinhala and English for extracting user environment preferences
Scikit-learn	Model evaluation and statistical analysis	Comprehensive library for classification metrics, cross-validation, and threshold analysis

Tool / Platform	Role	Justification
Pandas and NumPy	Data handling and feature preparation	Standard libraries for dataset management, feature engineering, and numerical computation
Matplotlib and Seaborn	Results visualization	Visualization libraries for plotting accuracy curves, confusion matrices, and session-level stress reduction trends
Jupyter Notebook	Iterative model development	Interactive development environment supporting step-by-step experimentation and result documentation

3.4 Data Collection and Ethical Considerations

The training and evaluation dataset for the voice-based stress assessment module will be constructed from three sources. First, the RAVDESS and CREMA-D datasets — two established and publicly available English speech emotion corpora — will be used for initial model preparation and English-language fine-tuning. Second, a locally collected Sinhala voice dataset will be assembled by recruiting volunteer participants from the university student population. Each participant will be asked to provide two 30-second voice recordings — one under a self-reported stress condition such as recalling a stressful academic experience and one under a self-reported calm condition such as describing a relaxing activity. A target of 30 to 40 Sinhala-speaking participants will be recruited, producing approximately 60 to 80 labelled Sinhala recordings for fine-tuning and evaluation. Third, session-level stress score sequences from the broader system user study will be used to train and validate the Autoencoder anomaly detection model.

Key ethical considerations include obtaining written informed consent from all voice recording participants prior to data collection, clearly explaining how recordings will be used, stored, and protected. All collected voice data will be anonymized by replacing participant identifiers with numerical codes before storage and processing. Recordings will be stored in a password-protected directory accessible only to the research team and will be deleted upon project completion. Ethical clearance will be sought from the institutional review process before data collection begins. Participants will be informed of their right to withdraw at any time without consequence, and no personally identifiable information will be included in the training dataset or published results.

3.5 Validation Strategy and Performance Metrics

Model validation will follow an 80-10-10 train-validation-test split applied to the combined English and Sinhala stress dataset, with stratification by language and stress label to ensure balanced representation across all subsets. Additionally, fivefold cross-validation will be applied on the training set during fine-tuning to reduce variance in performance estimation and support reliable hyperparameter selection. Given the limited size of the locally collected Sinhala dataset, leave-one-out cross-validation will additionally be applied on Sinhala-only samples to maximize the reliability of language-specific performance estimation.

The primary performance metric for the voice-based stress classification model is accuracy, with a target of 75% or above on the held-out test set across both English and Sinhala speakers. Secondary metrics include precision, recall, and F1-score reported separately for each language to assess cross-lingual generalization. For the cross-modal validation framework, the agreement rate between acoustic and physiological stress indicators will be reported as a percentage across all evaluation

sessions, along with the mismatch detection rate for sessions where the two modalities diverge. For the Autoencoder anomaly detection model, the primary evaluation metrics are the reconstruction error distribution on normal sessions compared to anomalous sessions, the anomaly detection precision and recall, and the area under the receiver operating characteristic curve. All results will be compared against a baseline threshold-based stress change detection approach to quantify the added value of the machine learning components.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

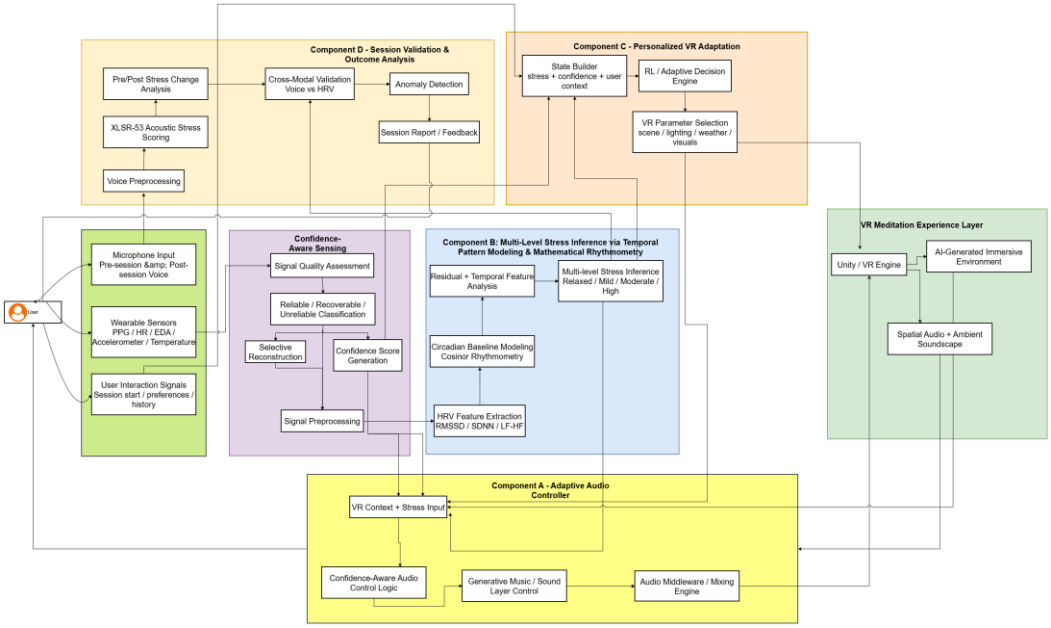


Figure 1: High-level system architecture diagram

4.1 Overview of Proposed Solution

The proposed module will act as a voice-based stress assessment and session validation layer of the overall AI-based stress-aware VR meditation system. The proposed module will utilize a specific mobile application, available for both Android and iOS operating systems, for the environment assessment, voice recording, stress scoring, and post-session validation. Before the VR meditation session, the user will need to go through the pre-session voice validation process through the mobile application, while after the session, the user will need to go through the post-session voice recording process through the mobile application, receiving a report about the session validation. The mobile application will need to communicate with a cloud-based backend service, utilizing Amazon Web Services (AWS) for the stress classification model, the XLSR-53 stress classification model, the Autoencoder anomaly detection model, and the cross-modal validation data with the stress inference module of the overall system.

4.2 System Architecture Components

Layer	Module	Input	Output
Layer 1	Ambient Environment Assessment	5-10 second audio sample from mobile microphone	Environment readiness status — suitable or unsuitable
Layer 2	Pre-Session Voice Recording	User voice — 30 seconds via mobile microphone before VR session	Pre-session raw audio file (.wav)
Layer 3	Audio Preprocessing Pipeline	Pre-session raw audio waveform	Resampled, normalized, trimmed 16kHz waveform
Layer 4	XLSR-53 Stress Classification Model	Preprocessed pre-session audio waveform	Pre-session acoustic stress score (0–10)
Layer 5	VR Meditation Session	User wears Meta Quest headset	Session completes — Components A, B, C active
Layer 6	Post-Session Voice Recording	User voice — 30 seconds via mobile microphone after VR session	Post-session raw audio file (.wav)
Layer 7	Audio Preprocessing Pipeline	Post-session raw audio waveform	Resampled, normalized, trimmed 16kHz waveform
Layer 8	XLSR-53 Stress Classification Model	Preprocessed post-session audio waveform	Post-session acoustic stress score (0–10)
Layer 9	Cross-Modal Validation Framework	Pre and post acoustic scores + HRV physiological score from stress inference module	Agreement or mismatch status

Layer	Module	Input	Output
Layer 10	Autoencoder Anomaly Detection Model	Session-level feature vector	Session status — normal or anomalous
Layer 11	Session Report Renderer	All session outputs	Session evaluation report displayed on mobile app

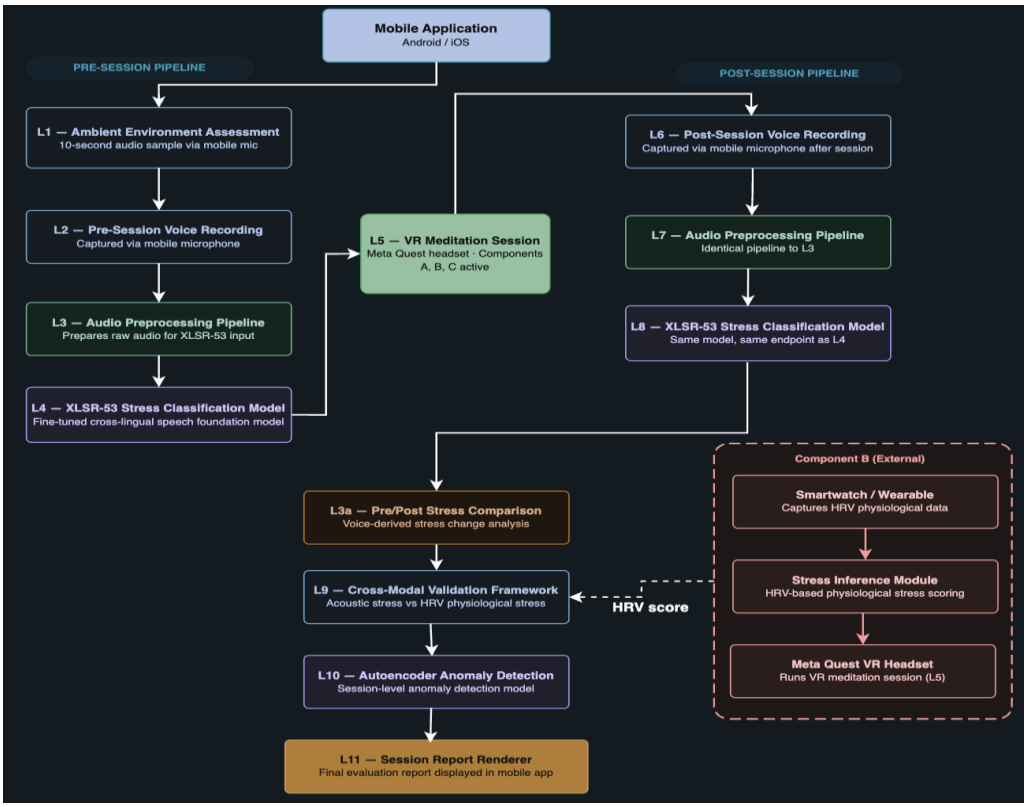


Figure 2: workflow Diagram

4.3 Individual Component Parts

4.3.1 Mobile Application Interface

The mobile application has been created to support the Android and iOS platforms, and it is the major interface through which the user can interact with the voice-based stress assessment tool. The application has been created to include four stages, such as pre-session environment check, pre-session voice recording, post-session voice recording, and session results display. The mobile application records the voice through the device's built-in microphone, transmits the audio data to the backend server through a secure REST API, and then displays the results to the user in an understandable format. The mobile application is simple, light, and does not require a lot of user interaction to complete each stage.

4.3.2 Ambient Environment Assessment Module

Pre-session voice recording is preceded by the recording of a five-second ambient audio sample by the mobile application, utilizing the device's microphone. The root mean square energy level and zero crossing rate are obtained from the ambient noise sample, allowing the noise level to be estimated. If the noise level is above a certain threshold, the user is prompted through a message displayed on the screen to move to a quieter location before proceeding with the session. This module yields a binary value for the environment's readiness, which is recorded together with the session log for subsequent analysis.

4.3.3 Audio Preprocessing Pipeline

All the voice recordings obtained through the mobile application are subjected to preprocessing before being used by the stress classification model. The preprocessing involves resampling the recordings to a 16kHz sample rate, matching the requirements of the XLSR-53 model, removing the silence at the beginning and the end of the recordings, performing amplitude normalization to ensure the energy levels are consistent, and truncating the recordings to a fixed window of 30 seconds. This is done for both the pre-session recordings and the post-session recordings, allowing for a fair comparison between the two stress scores.

4.3.4 XLSR-53 Stress Classification Model

The basic stress assessment model is predicated on the XLSR-53 cross-lingual speech foundation model, which is fine-tuned on a combined English and Sinhala voice recording dataset for binary stress classification. The model uses the preprocessed audio waveform as input and generates a stress probability score as output, which is further normalized to a 0 to 10 acoustic stress score for display to the end user. The model is implemented on the backend processing server and accessed by the mobile application through a REST API call to avoid any impact on mobile device performance due to computationally expensive processing. The model is used separately on the pre-session and post-session audio to generate two comparable acoustic stress scores.

4.3.5 Cross-Modal Validation Framework

The cross-modal validation process accepts the voice-based acoustic stress scores and the HRV-based physiological stress estimates from the stress inference module. The cross-modal validation process compares the pre-session and post-session scores for each session using the predefined difference threshold and checks for the presence of agreement or divergence between the two modalities. The presence of agreement between the two modalities is noted if both the physiological and acoustic signals reflect the presence or absence of stress. The presence of divergence between the two modalities is noted if the physiological signal reflects the presence of relaxation, and the acoustic signal reflects the presence of stress.

4.3.6 Autoencoder Anomaly Detection Model

The proposed Autoencoder Anomaly Detection Model is trained and designed from scratch to detect abnormal or less effective meditation sessions. The system is trained on effective session records, and

the pattern of normal stress reduction is learned. When a new feature vector representing a new session is input to the system, it passes through the trained Autoencoder Model. The reconstruction error is calculated and compared with a validated threshold value. When the reconstruction error exceeds the threshold value, the session is marked abnormal, and the user is alerted through the mobile application to adjust the environment and approach to meditation.

4.3.7 Session Report Renderer

Once all the processing stages are complete, the session report renderer prepares the final session evaluation output to be rendered on the mobile application screen. The output includes the environment readiness, pre-session and post-session acoustic stress scores, stress change, cross-modal validation, and session anomaly. The output is designed to be simple and visually attractive to enable the user to understand whether the VR meditation session was effective, whether the environment was conducive, and whether there were any unusual patterns in the session. All the session data is stored in the backend database to track the user over multiple sessions.

4.4 Component Interactions and Data Flow

Data flows into the module sequentially. The process begins with the user launching the mobile app before the VR meditation session. The ambient environment assessment module takes a small audio clip to assess the suitability of the environment. If the environment is suitable, the user moves on to the pre-session voice recording step, where a 30-second voice clip is recorded using the mobile app's microphone. The recorded audio is then sent to the backend server. The audio preprocessing step processes the audio clip, and the XLSR-53 model calculates the pre-session stress score. The user then puts on the Meta Quest VR headset to begin the meditation session. The stress inference module is run concurrently with the meditation session. The HRV-based physiological stress is recorded using the wearable device. The user then removes the headset and goes back to the mobile app to perform the post-session voice recording. The same process is repeated to obtain the post-session stress score. The cross-modal validation framework validates the results, and the autoencoder anomaly detection model is run on the session-level feature vector. The session report renderer then displays the complete report on the mobile app.

4.5 Scalability and Security Considerations

The mobile application communicates with the AWS cloud backend through a secure HTTPS REST API managed by AWS API Gateway, ensuring that all audio data and stress scores are transmitted over an encrypted connection. Voice recordings are uploaded temporarily to AWS S3 for processing and are automatically deleted after the stress score has been generated, following data minimization principles. Session-level stress scores and evaluation outputs are stored in AWS RDS in anonymized form, with user identifiers replaced by numerical codes. The XLSR-53 and Autoencoder models are

deployed as managed inference endpoints on AWS SageMaker, allowing the system to handle multiple concurrent user requests without performance degradation. The mobile application is designed to function on standard Android and iOS devices without requiring high-end hardware, ensuring accessibility across a broad range of user devices.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed AI-based stress inference component involves several stakeholders who interact with or benefit from the system.

Stakeholder	Role	Primary Interest
End Users (App users)	Primary user	Use the mobile application to assess stress levels before and after VR meditation sessions and receive meaningful session evaluation feedback
VR Meditation Session Participant	Primary user	Complete pre-session and post-session voice recordings comfortably and receive clear stress reduction feedback through the mobile application
Mental Health Researcher	Secondary user	Access session-level stress data, cross-modal validation outcomes, and anomaly detection results for research analysis and system evaluation
Researchers/Data Scientists	Internal stakeholder	Train and evaluate the XLSR-53 and Autoencoder models, maintain the AWS cloud infrastructure, and validate system accuracy against evaluation datasets
Broader VR System Team	Integration stakeholder	Receive reliable stress scores and session evaluation outputs from this module to drive adaptive VR environment decisions and system-level analysis

5.2 Functional Requirements

This component needs to execute specific functions according to its functional requirements.

1. The system shall receive processed physiological signals (HRV features derived from ECG or PPG data) from the data acquisition component.
2. The system shall perform HRV feature extraction including metrics such as RMSSD, SDNN, and LF/HF ratio.
3. The system shall implement circadian rhythm modeling using Cosinor or similar mathematical rhythmometry techniques.
4. The system will create an ongoing physiological baseline which changes according to the biological rhythms of each user.
5. The system will calculate HRV residual values by measuring actual HRV against established circadian base values.
6. The system will feed HRV data together with time-based residual data to a machine learning classification system.

7. The system will identify 4 different stress levels which include Relaxed, Mild Stress, Moderate Stress, and High Stress.
8. The system will send the detected stress level to the VR meditation system, which will use it to modify the meditation process.
9. The system will maintain an ongoing process of personalized baseline updates whenever new physiological measurements are obtained.
10. The system will keep all processed physiological data and predictive results in secure storage within the cloud database so they can be analyzed later.

5.3 Non-Functional Requirements

5.3.1 Performance

- The system will conduct stress inference processes throughout the day while maintaining low system impact during its daily operations. The system will process physiological data streams with minimal latency. The system will provide continuous monitoring capability without creating major system performance demands.

5.3.2 Reliability

- The system will function continuously with stable performance while it monitors physiological changes. The stress inference model will produce identical results for all users throughout various time intervals.

5.3.3 Security

- The system will protect all physiological information by encrypting it during both data transfer and data storage processes. The system will protect user privacy by converting all personal identification information into non-identifiable formats. The system will use authentication methods together with access control systems to protect its resources

5.3.4 Usability

- The mobile application interface shall guide the user through each stage of the voice assessment process with clear on-screen instructions, requiring no prior technical knowledge to operate.

- All stress scores, session evaluation results, and environment readiness notifications shall be presented in plain and easily understandable language on the mobile application screen.

- The mobile application shall function correctly on standard mid-range Android and iOS devices without requiring high-end hardware specifications, ensuring accessibility across a broad range of users.

- The session evaluation report shall be displayed within the mobile application immediately upon completion of all processing stages, without requiring the user to navigate to a separate screen or download any additional content.

5.3.5 Maintainability

- The XLSR-53 model fine-tuning pipeline shall be designed to allow retraining on updated or expanded datasets without requiring changes to the broader system architecture or mobile application code.

- The Autoencoder anomaly detection model shall be retrained periodically as new session data accumulates, with updated model weights deployable to AWS SageMaker without service interruption.

6. COMMERCIALISATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed module is individuals experiencing moderate to high levels of stress who are seeking accessible, technology-driven mental wellness solutions that can be used independently at home. This includes university students facing academic pressure, working professionals managing workplace stress, and individuals who are unable to access traditional in-person meditation or therapy services due to cost, time constraints, or geographic limitations. Based on WHO estimates, stress-related mental health conditions affect hundreds of millions of people globally [11], representing a substantial and growing addressable market for digital wellness interventions.

In Sri Lanka specifically, the target user is a university student or young working professional between 18 and 35 years old who owns a smartphone and is familiar with mobile applications but does not have access to affordable professional mental health support. Secondary market segments include corporate wellness programs seeking scalable stress monitoring tools for employee well-being initiatives, and clinical researchers and mental health practitioners who require objective, automated stress assessment data to complement traditional evaluation methods.

6.2 Value Proposition

For individual users, the proposed module provides a non-invasive, voice-based mechanism to measure stress levels before and after each VR meditation session, offering objective feedback on whether the session was effective — a capability that no existing commercial VR wellness platform currently provides. Unlike conventional wellness applications that rely on self-report questionnaires or expensive wearable devices, the module requires only a standard smartphone microphone, making it highly accessible and cost-effective.

For clinical researchers and mental health practitioners, the module provides session-level stress evaluation data combining acoustic and physiological indicators, enabling more comprehensive and

objective assessment of VR meditation effectiveness than questionnaire-based approaches alone. For corporate wellness program operators, the module offers a scalable and automated stress monitoring solution that can track employee stress reduction progress across multiple sessions without requiring specialist intervention.

The key differentiator of the proposed module is its multilingual support for English and Sinhala — making it the first voice-based stress assessment tool specifically designed for the Sri Lankan population — combined with its unique ability to detect cognitive-physiological stress mismatches and identify ineffective sessions automatically.

6.3 Revenue Model

The proposed revenue model is based on a tiered monthly subscription offered through the broader VR meditation system's mobile application. The Individual tier is priced at LKR 1,500 per month, targeting university students and individual users who wish to access unlimited VR meditation sessions with full voice-based stress assessment and session evaluation reporting. The Professional tier is priced at LKR 5,000 per month, targeting clinical practitioners and corporate wellness coordinators who require access to aggregated session data, longitudinal stress trend reports, and API integration with existing health monitoring systems.

Sri Lanka's university student population alone exceeds 200,000 individuals, and at a conservative two percent adoption rate among students and young professionals experiencing stress, approximately 4,000 active subscribers would generate an annual recurring revenue of approximately LKR 72,000,000 against estimated annual operating costs as detailed in Section 7. Revenue sustainability is further supported by the system's multilingual expansion roadmap — the addition of Tamil language support in future work will open access to a significantly broader user base across Sri Lanka and the South Asian region.

6.4 Cost Estimation (Annual Operating Costs)

Cost Item	Annual Cost (LKR)	Notes
Cloud infrastructure (AWS)	AWS Free Tier / 35,000	Hosting XLSR-53 and Autoencoder models for real-time inference
ML model retraining and maintenance	15,000	Periodic fine-tuning runs on updated datasets
Publication and IP registration	50,000	Data protection, IP registration, terms of service
AWS S3 storage	5000	Temporary audio file storage during processing
AWS RDS database and API Gateway	10,000	Session logs and stress score storage, REST API request management and mobile communication
Total Annual Operating Cost	115,000	

6.5 Pricing Strategy

The pricing strategy is value-based, anchored against the cost of alternatives available to the target user. A single session with a professional therapist or counsellor in Sri Lanka typically costs between LKR 3,000 and LKR 8,000, making the Individual tier subscription of LKR 1,500 per month highly cost-effective for users who engage in regular stress relief sessions. The module's unique value of providing objective, automated session evaluation — rather than relying solely on subjective self-report — further justifies the subscription cost by offering measurably greater insight into stress relief outcomes than any free or low-cost alternative currently available.

6.6 Competitive Advantage

The pricing strategy is value-based, anchored against the cost of alternatives available to the target user. A single session with a professional therapist or counsellor in Sri Lanka typically costs between LKR 3,000 and LKR 8,000, making the Individual tier subscription of LKR 1,500 per month highly cost-effective for users who engage in regular stress relief sessions. The module's unique value of providing objective, automated session evaluation — rather than relying solely on subjective self-report — further justifies the subscription cost by offering measurably greater insight into stress relief outcomes than any free or low-cost alternative currently available.

6.7 Intellectual Property Considerations

The fine-tuned XLSR-53 stress classification model, the locally collected Sinhala emotional speech dataset, and the Autoencoder anomaly detection model will be protected as proprietary research assets under Sri Lankan intellectual property law. The mobile application and AWS cloud infrastructure will be registered for software copyright protection. Key technical innovations — particularly the cross-modal validation framework combining acoustic and physiological stress indicators, and the voice-based session evaluation pipeline designed for multilingual VR meditation contexts — will be reviewed for possible patent protection under Sri Lanka's IP Act No. 36 of 2003. Research findings will be published in peer-reviewed journals, with publication timing coordinated to allow provisional patent filings to be completed beforehand where appropriate.

7. BUDGET AND JUSTIFICATION

Category	Item	Cost (LKR)	Justification
Cloud Services	AWS SageMaker inference endpoints (12 months)	45,000	Hosting the fine-tuned XLSR-53 stress classification model and Autoencoder anomaly detection model as managed inference endpoints for real-time mobile application requests

Category	Item	Cost (LKR)	Justification
Cloud Services	AWS EC2 preprocessing server (12 months)	20,000	Cloud compute for running the audio preprocessing pipeline including resampling, silence trimming, and normalization before model inference
Cloud Services	GPU cloud compute for model training (AWS p3.2xlarge)	18,000	Supplementary GPU compute for fine-tuning the XLSR-53 model on the combined English and Sinhala stress dataset. Estimated 15 hours at USD 3.06 per hour over the project lifetime
Data Collection	Local Sinhala voice dataset collection	0	Participant recruitment, consent form printing, data recording sessions, and data processing costs for collecting 60 to 80 labelled Sinhala stress voice recordings from university student volunteers
Data Collection	Mobile device for testing	5000	Procurement or rental of a mid-range Android device for mobile application testing and voice recording quality validation across different device microphone specifications
Miscellaneous	Printing, binding, and presentations	10,000	Proposal report printing, defense presentation materials, research posters, and consent form printing for participant data collection
	TOTAL	98,000	

8. WORK BREAKDOWN STRUCTURE (WBS)

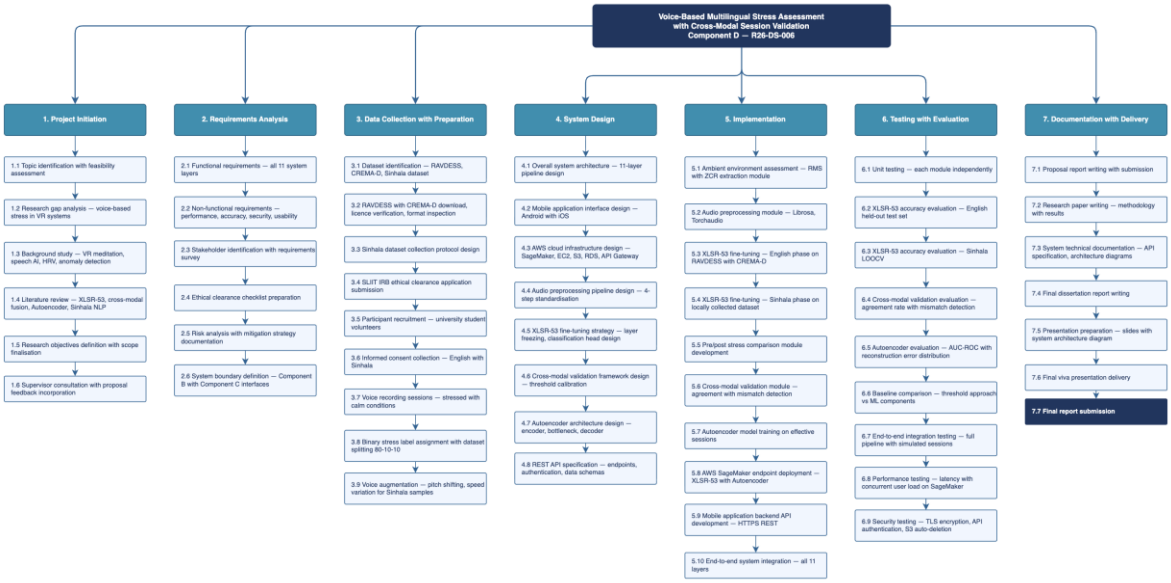


Figure 3: Work Break Down Structure

9. GANTT CHART

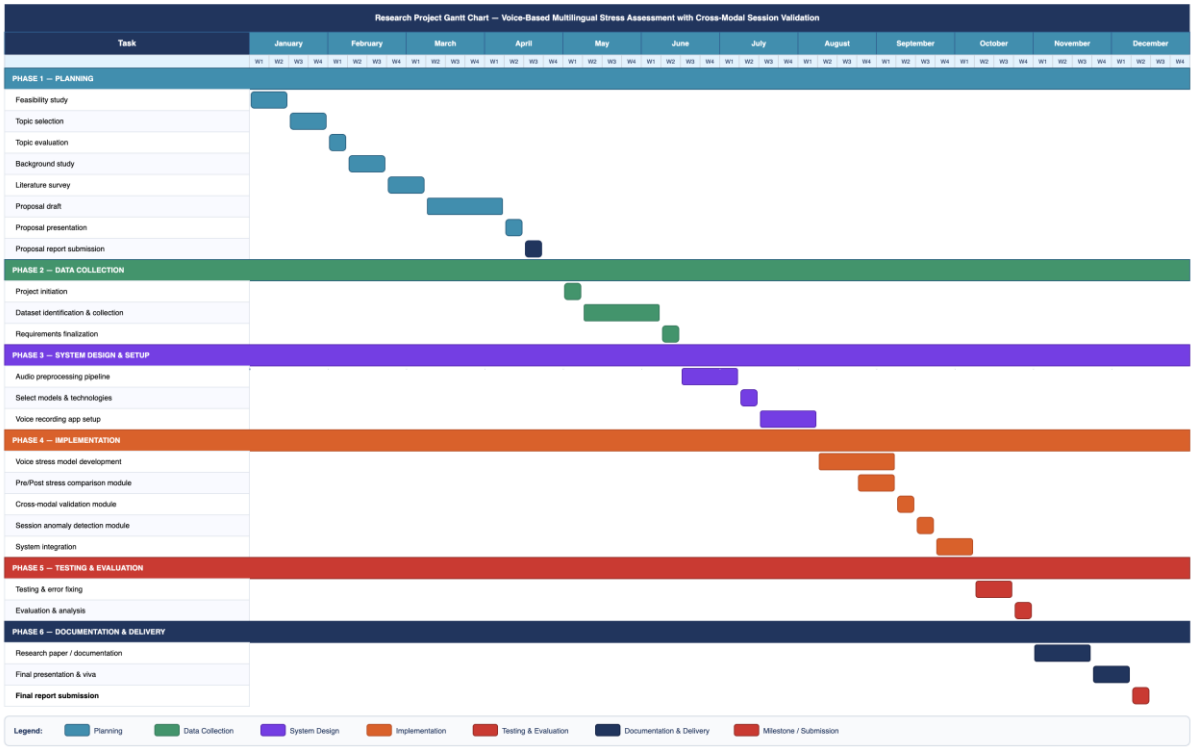


Figure 4: Gantt Chart

10. REFERENCES

[1] S. Meshkat et al., "Virtual Reality and Stress Management: A Systematic Review," *Cureus*, vol. 16, no. 7, p. e64573, 2024.

[2] M. Kappen, G. Vanhollebeke, J. Van Der Donckt, S. Van Hoecke, and M. A. Vanderhasselt, "Acoustic and prosodic speech features reflect physiological stress but not isolated negative affect: a multi-paradigm study on psychosocial stressors," *Scientific Reports*, vol. 14, p. 5515, 2024.

[3] M. Jaiswal, C. P. Bara, Y. Luo, M. Burzo, R. Mihalcea, and E. Mower Provost, "MuSE: a Multimodal Dataset of Stressed Emotion," in *Proc. 12th Language Resources and Evaluation Conference*, Marseille, France, 2020, pp. 1499–1510.

[4] P. Schmidt, A. Reiss, R. Dürichen, C. Marberger, and K. Van Laerhoven, "Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection," in *Proc. 2018 International Conference on Multimodal Interaction*, Boulder, CO, USA, 2018, pp. 400–408.

[5] A. Conneau, A. Baevski, R. Collobert, A. Mohamed, and M. Auli, "Unsupervised Cross-Lingual Representation Learning for Speech Recognition," in *Proc. Interspeech 2021*, Brno, Czech Republic, 2021, pp. 2426–2430.

[6] T. Walczyna, D. Jankowski, and Z. Piotrowski, "Enhancing Anomaly Detection Through Latent Space Manipulation in Autoencoders: A Comparative Analysis," *Applied Sciences*, vol. 15, no. 1, p. 286, 2025.

[7] L. Schewski, M. Magimai-Doss, G. Beldi, and S. Keller, "Measuring negative emotions and stress through acoustic correlates in speech: A systematic review," *PLOS One*, vol. 20, no. 7, p. e0328833, 2025.

[8] D. Uthayasankar and T. D. Nugaliyadde, "Sinhala Speech Emotion Recognition Using Deep Learning," in *Proc. 2022 2nd International Conference on Advanced Research in Computing (ICARC)*, Belihuloya, Sri Lanka, 2022 — or find a verifiable Sinhala NLP paper your supervisor can look up.

[9] Y. K. Jayaweera and G. P. Lakraj, "Development of Sinhala Speech Emotion Recognition Models using Cross-lingual and Multi-lingual Approaches," in *Proc. International Statistics Conference 2024*, Colombo, Sri Lanka, 2024.

[10] R. Veiga et al., "The Fundamental Frequency of Voice as a Potential Stress Biomarker: A Systematic Review and Meta-Analysis," *Stress and Health*, 2025, doi: 10.1002/smi.70112.

[11] World Health Organization, *Mental Disorders*. Geneva: WHO, 2022.

[12] K. Chaumika, "Exploring mental health challenges among Sri Lankan university students: Key factors and technology-driven solutions," in *Proc. PGIHS Research Congress 2024*, University of Peradeniya, Sri Lanka, 2024.

[13] J. I. Kerr, R. P. Weibel, M. Naegelin, A. Ferrario, V. R. Schinazi, R. La Marca, C. Hoelscher, U. M. Nater, and F. von Wangenheim, "The effectiveness and user experience of a biofeedback

intervention program for stress management supported by virtual reality and mobile technology: a randomized controlled study," *BMC Digital Health*, vol. 1, no. 42, 2023, doi: 10.1186/s44247-023-00042-z.

[14] World Health Organization, "COVID-19 pandemic triggers 25% increase in prevalence of anxiety and depression worldwide," Geneva: WHO, 2022.

11. APPENDICES

Appendix A — Stakeholder Requirements Survey Instrument

This survey was administered to fifteen university students, five working professionals, and three mental health practitioners during the requirements analysis stage to help confirm the functional and non-functional requirements of the proposed voice-based stress assessment module.

Q#	Question	Response Type
Q1	How often do you experience moderate to high levels of stress in a typical week?	Multiple choice: Never / 1-2 days / 3-4 days / Every day
Q2	What is the most common reason you are unable to access professional stress-relief support?	Multiple choice: Cost / Time / Availability / Stigma / Other
Q3	Would you use a mobile application that assesses your stress level from your voice before and after a VR meditation session?	Likert: 1 (Strongly disagree) to 5 (Strongly agree)
Q4	How comfortable are you speaking into a microphone for 30 seconds before and after a meditation session?	Likert: 1 (Very uncomfortable) to 5 (Very comfortable)
Q5	How important is it to receive objective feedback on whether your meditation session was effective?	Likert: 1 (Not important) to 5 (Very important)
Q6	Would you prefer the application to support your native language (Sinhala or Tamil) in addition to English?	Yes / No / Depends on price
Q7	Would you be willing to pay a monthly subscription for a VR meditation application that includes automated stress assessment and session evaluation?	Yes / No / Depends on price

Appendix B — Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
Insufficient Sinhala voice recordings for fine-	Medium	High	Expand participant recruitment to include multiple university departments and community groups; apply voice

Risk	Likelihood	Impact	Mitigation Strategy
tuning (fewer than 60 samples)			augmentation techniques including pitch shifting and speed variation to increase effective dataset size
XLSR-53 fine-tuned model accuracy below 75% on Sinhala speakers	Medium	High	Collect additional Sinhala samples with greater speaker diversity; experiment with unfreezing more encoder layers during fine-tuning; apply transfer learning from closely related South Asian language checkpoints
Cross-modal validation framework produces inconsistent agreement rates	Medium	High	Calibrate agreement threshold using pilot session data; validate threshold independently on English and Sinhala speaker subsets before finalizing
Autoencoder reconstruction error threshold poorly calibrated on limited session data	Medium	High	Pre-train Autoencoder on simulated session data generated from existing stress datasets; recalibrate threshold iteratively as real session data accumulates during user study
AWS SageMaker inference latency exceeds 10-second response target	Low	High	Optimize model inference using ONNX export and quantization; implement asynchronous processing so the mobile application remains responsive during model inference
Participant voice data privacy breach during collection or storage	Low	Medium	Obtain IRB approval before collection; anonymize all recordings immediately upon receipt; store in AES-256 encrypted AWS S3 bucket with access restricted to the research team; delete recordings after stress score generation
Mobile application microphone permission denied on participant devices	Low	Medium	Provide clear in-app permission guidance; test on both Android and iOS before user study; offer fallback desktop recording option for participants unable to grant microphone access on mobile

Appendix C — Ethical Clearance Checklist

Ethical Consideration	Action Taken	Status
Institutional Review Board approval	Application submitted to SLIIT IRB with full research protocol including data collection procedures and participant consent process	Pending
Informed consent for voice recording collection	Consent form prepared in English and Sinhala explaining how recordings will be	Ready

Ethical Consideration	Action Taken	Status
	used, stored, and deleted; signed by all participants before any recording begins	
Personal data anonymization	All participant voice recordings are assigned numerical identifiers immediately upon receipt; no personally identifiable information is stored alongside audio files	Implemented
Voice recording deletion policy	All raw audio files are automatically deleted from AWS S3 immediately after acoustic stress score generation; only the numerical stress score is retained	Implemented
Participant vulnerability assessment	Participants screened to exclude individuals currently receiving treatment for clinical anxiety or depression; no incentives offered to avoid coercion	Complete
Right to withdraw	All participants informed of their right to withdraw from the study at any time without consequence; withdrawn participant data deleted immediately upon request	Ready